

ENGINEER'S COMPONENT HANDBOOK

Every component, why it is there, and how it all connects.

Large-print edition · scan any QR for the manufacturer page

Rev K design basis · 375 kW / 1,221 kWh · ERCOT

T H E S I T E I N T H R E E L I N E S

3x CATL EnerOne+ battery cabinets (407 kWh each, liquid-cooled).

1x Kehua Cabinet-375 PCS: 3x 125 kW inverters + built-in 30 kVA aux transformer.

1x control panel (this document) - the brain. Built by STR Medical.

P E D I G R E E + V E R I F I C A T I O N

Consolidates two field-proven Carson, CA panels. Every CATL pin confirmed

against CATL's factory checklist. Automated tests re-verify all 174 circuits

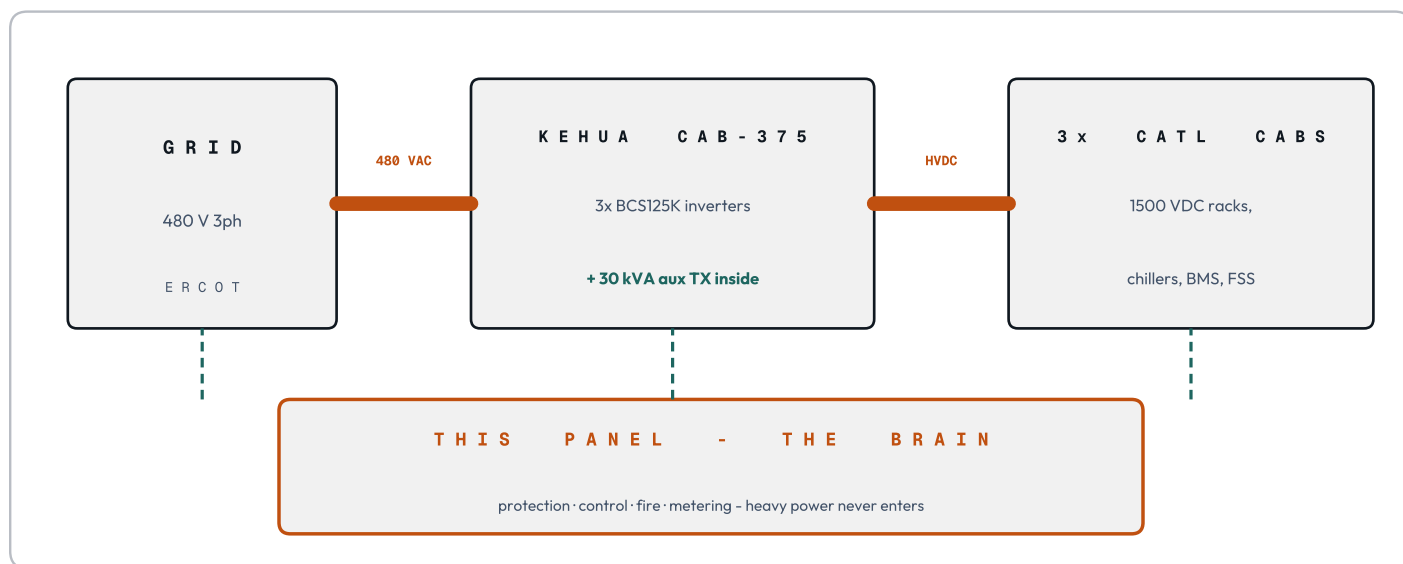
after any change. Two independent reviews (15-agent swarm, then a six-

discipline PE-rigor board) - every drawable finding fixed in Rev K.

S T A N D I N G R U L E

Nothing here is released for construction. PE stamp + page 19 gates come first.

THE 60-SECOND MAP



THREE SEPARATE POWER SYSTEMS - MEMORIZE THIS

MAIN 480 V

Grid <-> PCS <-> batteries, up to 375 kW.

Only a 40 A housekeeping tap (Q1) enters this panel.

AUX 380Y / 220 V

From the TX inside the Kehua cabinet -> Q2 -> CB3-CB8 ->

chillers + rack BMS. Two-source panel: NEC 409.110 label.

24 VDC

Made in-panel, twice redundant. Rail A: PSU1/2 + UPS (EMS, net,

IO, fire trip). Rail B: PSU3/4 (MBMUs, ETH). Health contacts to EMS.

Control hierarchy: EMS (CPU1/2) > MBMU x3 (this panel) > SBMU (per rack) > CSC x8 (per rack).

THE DOCUMENT SET

WHERE TO LOOK, IN ORDER

1

Z1-Sugarland.kicad_sch + PDF (one-page ARCH E)

THE master schematic - every wire on one sheet. Start here.

2

Live Simulator (html)

Animated model, six scenarios, slow-motion. Best 20 minutes of onboarding.

3

RevJ swarm findings (json)

The independent review: what was found, what changed. Read before 'fixing' anything.

4

BOM Rev K (xlsx)

Every model number in this handbook, with prices and sourcing notes.

5

8-sheet A3 set / Altium SchDoc

Same netlist by discipline / native Altium copy with part parameters.

6

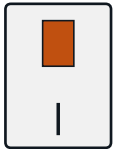
CATL + Kehua factory docs

External wiring diagram, checklist, specs, Modbus maps - in the project archive.

Rule of thumb: if this handbook and the schematic ever disagree, the schematic wins -

then tell engineering, because the handbook is generated from the same database and should never disagree.

480 V POWER TRAIN (1 OF 2)



Q1

40A 3P UL489 (TMC S3D class)

P H O E N I X C O N T A C T

phoenixcontact.com

PURPOSE

Panel main disconnect for the 480 V housekeeping feed.

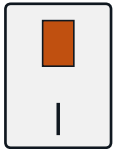
WHY THIS ONE

Small feed only - heavy power stays outside.

40 A covers CPT + SPD + meter taps.



SCAN : D A T A S H E E T



CB1 + SPD1

S203-K30 + Type 2 SPD 480/277Y

A B B + R A Y C A P / E A T O N

raycap.com

PURPOSE

Surge protection for the whole panel; CB1 is its disconnect.

WHY THIS ONE

Gulf-coast lightning. SPD leads under 0.5 m per drawing note. Verify In/AIC at IFC.



SCAN : D A T A S H E E T



F1-F5

KLKR Class CC + holders

L I T T L E F U S E + A L T E C H

littelfuse.com

PURPOSE

F1-F3: 1 A meter voltage taps. F4-F5: CPT primary.

WHY THIS ONE

Class CC = 200 kA interrupting rating in a DIN footprint. Holder: Altech 2540104, Carson-proven.



SCAN : D A T A S H E E T

480 V POWER TRAIN (2 OF 2)



T2 (CPT)
500 VA 480>120/240
encapsulated

S O L A H D (E 5 0 0 C L A S S)

solahd.com

PURPOSE

Makes 120/240 V control power from the 480

V feed.

WHY THIS ONE

Sized with margin for panel A/C + electronics.

N-G bond BD1 at its secondary (NEC 250.30).



SCAN : D A T A S H E E T



PT1
PT set 480:120, metering
class

F U N C T I O N A L D E V I C E S / I T

functionaldevices.com Metering-class accuracy. Exact ratio verified

against Carson at IFC.

PURPOSE

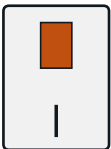
Steps 480 V down to the meter's voltage-

reference range.

WHY THIS ONE



SCAN : D A T A S H E E T



Q2 (REV J)
25A 3P 400V UL489

A B B / E A T O N

abb.com

PURPOSE

Incoming disconnect for the 380Y/220 V aux

feeder from the Kehua cabinet.

WHY THIS ONE

UL508A requires a disconnect per incoming

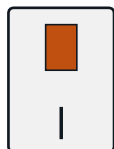
supply. Panel carries a TWO SOURCES label (NEC

409.110).



SCAN : D A T A S H E E T

AUX DISTRIBUTION TO THE CABINETS



CB3, CB5, CB7

20A 1P C-curve (REV J

UPSIZE)

A B B / E A T O N

abb.com

PURPOSE

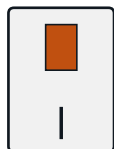
One chiller branch per CATL cabinet,
phases balanced L1/L2/L3.

WHY THIS ONE

CATL 3 kW continuous = $13.6 \text{ A} \times 125\% = 17 \text{ A}$,
so 15 A was too small. C-curve rides the 31 A
start surge. Branch wire AWG10.



SCAN : D A T A S H E E T



CB4, CB6, CB8

6A 1P C-curve UL489

A B B / E A T O N

abb.com

PURPOSE

Rack-BMS aux branch per cabinet (about 150
W each).

WHY THIS ONE

A dedicated small breaker per cabinet – a BMS
fault never darkens a neighbor's cooling.



SCAN : D A T A S H E E T



TB1, TB13, TB2-4

**UT 6 + UT 6-PE feed-
through**

P H O E N I X C O N T A C T

phoenixcontact.com

PURPOSE

Landing terminals: 480 V entry, aux feeder
in, cabinet circuits out.

WHY THIS ONE

UT series = insulated feed-through.
Green/yellow UT 6-PE only on the PE poles.



SCAN : D A T A S H E E T

24 VDC – THE LIFEblood (1 OF 2)

PSU1, PSU2



NDR-75-24 (75 W / 3.2 A)

M E A N W E L L

meanwell.com

PURPOSE

Redundant pair feeding the EMS computers and network.

WHY THIS ONE

Two supplies + a redundancy module: either one can fail without dropping the brain.



SCAN : D A T A S H E E T

RM1



DRDN20-24 redundancy module

M E A N W E L L

meanwell.com

PURPOSE

Diode-ORs PSU1 and PSU2 onto one 24 V bus.

WHY THIS ONE

True N+1 – a shorted supply cannot drag down the healthy one.



SCAN : D A T A S H E E T

UPS1



QUINT4-CAP/24DC/5/4KJ supercap

P H O E N I X C O N T A C T

phoenixcontact.com

PURPOSE

Rides the EMS/network 24 V bus through dips and outages.

WHY THIS ONE

Supercapacitors: no battery maintenance, full rating in a hot enclosure.



SCAN : D A T A S H E E T

24 VDC – THE LIFEBLOOD (2 OF 2)



PSU3 + PSU4 + RM2 (REV K)

NDR-240-24 pair + DRDN40 module

MEAN WELL

meanwell.com

PURPOSE

Redundant pair feeding MBMU x3 and ETH x3 through an ORing module.

WHY THIS ONE

Rev K fix: one supply alone used to carry all battery telemetry - now either can fail, and its DC-OK contact tells the EMS the moment it does.



SCAN: DATASHEET



F6-F12

KLKR, 120 V branch schedule

LITTELFUSE + ALTECH

littelfuse.com

PURPOSE

Branch fusing: PSU1-3, FACP AC, EMS CPU1-2, network.

WHY THIS ONE

Per-device fusing per Carson - one fault takes out one device, and the label says which.



SCAN: DATASHEET



F13-F17

KLKR, 24 V branches BMS1/2/3, IO, ETH

LITTELFUSE + ALTECH

littelfuse.com

PURPOSE

DC fusing for the CATL masters, ioLogik and ETH modules.

WHY THIS ONE

Fuse labels match Carson's as-built holder labels exactly - field techs already know them.



SCAN: DATASHEET

CONTROL + NETWORK (1 OF 2)

CPU1, CPU2



UC-2222A-T-US (Arm,
Linux)

M O X A

moxa.com

PURPOSE

Redundant EMS computers running the
Battalion Energy image.

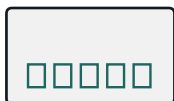
WHY THIS ONE

Fanless industrial DIN units, -40..70 C; two
so one can fail or be re-imaged live.



SCAN : DATASHEET

SW1



EDS-4014 managed switch,
14-port

M O X A

moxa.com

PURPOSE

The single network hub. VLAN 10 WAN / 20
control / 30 BMS.

WHY THIS ONE

One switch replaces Carson's three;
segregation by VLAN. Battalion signs off the
plan; config sheet at commissioning.



SCAN : DATASHEET

I01



ioLogik E1214 (6 DI / 6
DO)

M O X A

moxa.com

PURPOSE

Reads FACP ALARM/SUPV/TROUBLE (wired fail-
safe N/C) + supply health DI3-5.

WHY THIS ONE

The EMS sees fire state and the FIRST power-
supply failure - on the UPS rail so it
survives an outage (Rev K).



SCAN : DATASHEET

CONTROL + NETWORK (2 OF 2)

MBMU1-3, EM1-3



CATL master BMS + ETH
modules

C A T L (W I T H C A B I N E T S)

catl.com

PURPOSE

MBMU = battery master per cabinet (CAN).

ETH = its Ethernet translator.

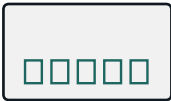
WHY THIS ONE

Powered from panel 24 V, NOT cabinet aux -
they survive an aux outage. Each ETH housing
bonds to PE (CATL checklist).



SCAN : D A T A S H E E T

LTE



Cellular backup router

C A R R I E R P E R C O N T R A C T

-

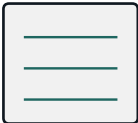
PURPOSE

Backup WAN path when site internet fails.

WHY THIS ONE

Operators never lose visibility; the site
itself runs fine without internet.

PCS LINK



Modbus TCP over Ethernet
(VLAN 20)

K E H U A P R O T O C O L D O C

kehua.com

PURPOSE

EMS commands the three 125 kW inverters:
mode, setpoints, status.

WHY THIS ONE

TCP over RS-485 = one cable plant, switch-
monitored. Register map per Kehua manual in
project docs.



SCAN : D A T A S H E E T

FIRE SYSTEM (1 OF 2)



FACP1
**ARC-100 addressable
releasing panel**

P O T T E R
pottersignal.com

PURPOSE
The fire brain: detection, siren/strobe,
releasing circuits, own batteries.

WHY THIS ONE
Releasing-listed, not just alarm. Its 49 C
ambient limit drives the enclosure cooling
spec.



SCAN : D A T A S H E E T



CA-6075 + PSN-1000
**Class A SLC card +
releasing PSU**

P O T T E R
pottersignal.com

PURPOSE
SLC rings rack-to-rack (Rev K); PSN-1000
powers all three aerosol circuits.

WHY THIS ONE
Class A survives a cut cable. CATL suppression
draws 1.14 A/cab in alarm vs ARC-100's 1
A/circuit - expect PSN needed; final at ICD.



SCAN : D A T A S H E E T



BAT1, BAT2
**PS-1280 12V 8Ah SLA
(pair = 24 V)**

P O W E R - S O N I C
power-sonic.com

PURPOSE
FACP standby + alarm power per NFPA 72.

WHY THIS ONE
Preliminary calc: 4.8 Ah required. Finalize
with measured quiescent current.



SCAN : D A T A S H E E T

FIRE SYSTEM (2 OF 2)



SIR1, STB1, K1

Siren + strobe + PCS
trip relay

E A T O N (S T R O B E)

eaton.com

PURPOSE

Warn people; give a trained person a last-second veto on release.

WHY THIS ONE

K1 is Rev K's most important part: held energized by the FACP, it drops on alarm/failure/power loss and stops the inverter by wire - no software in the path.



SCAN : D A T A S H E E T



R4 + EOLs, TB8-14
EOLD 3005012 x4 + UK 5 N
terminals

P O T T E R + P H O E N I X

pottersignal.com

PURPOSE

EOL devices supervise every circuit; UK 5 N = insulated feed-through.

WHY THIS ONE

8 EOLs now IN the netlist (Rev K): NAC, 3 aerosol circuits at the JX1 end, abort, manual release, 2 monitor inputs. Every supervised circuit really is supervised.



SCAN : D A T A S H E E T



M1
EG4115 (Core), 15 CT
inputs

E G A U G E

egaug.net

PURPOSE

Revenue metering of Pol, solar and BESS - three phases each.

WHY THIS ONE

Single meter replaces Carson's several (approved deviation). Open-circuit-safe CT inputs.



SCAN : D A T A S H E E T

METERING + ENCLOSURE



CT1-9 + TB11

ECS split-core 333 mV +

UT 4

E GAUGE + P HOENIX

egauge.net

PURPOSE

Current sensing on nine conductors; TB11

lands the 18 AWG twisted leads.

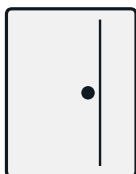
WHY THIS ONE

333 mV CTs cannot hurt anyone if unplugged -

no shorting blocks needed.



SCAN: DATASHEET



ENCLOSURE

NEMA 4, ~72 x 48 x 24 in

H OFFMAN / S AGINAW

hoffman.nvent.com

PURPOSE

Houses everything on these drawings;

outdoor pad mount.

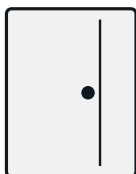
WHY THIS ONE

Scope requires NEMA 4 minimum - fan-vented 3R

is PROHIBITED. Final size at layout.



SCAN: DATASHEET



A/C + HEATER

1500 BTU/h A/C + 100 W

Stego

P FANNENBERG + S TEGO

pfannenberg.com

PURPOSE

A/C keeps the FACP under its 49 C limit;

heater stops condensation.

WHY THIS ONE

Heat load fell to ~99 W once the aux TX left

the enclosure. Select A/C at IFC.



SCAN: DATASHEET

THE TERMINAL RULE + POWER SOURCE

TERMINAL RULE



UT/UK feed-through +

USLKG 5 x3

PHOENIX CONTACT

phoenixcontact.com

PURPOSE

Insulated feed-through terminals for ALL circuits, everywhere.

WHY THIS ONE

USLKG (green/yellow, rail-bonded) ONLY at CAN shield poles SB1-3. Anywhere else = the build is wrong. STOP IT. (Rev J critical fix.)



SCAN: DATASHEET

SB1-SB3



USLKG 5 - CAN shield
ground poles

PHOENIX CONTACT

phoenixcontact.com

PURPOSE

The single-point shield ground, one per cabinet's CAN run.

WHY THIS ONE

Shield grounds at the panel end only and floats at the cabinet - CATL requirement, now modeled in the netlist.



SCAN: DATASHEET

X18 (FIELD)



Kehua 30 kVA 480/380 aux
TX

KEHUA (IN PCS CABINET)

kehua.com

PURPOSE

The aux power source: transformer + 3 output breakers in the PCS cabinet.

WHY THIS ONE

Kehua-included (verified in their spec) - replaced a \$6.9k custom pad TX. The aux N-G bond lives at THIS transformer, never in our panel.



SCAN: DATASHEET

CATL PINS THAT MUST BE RIGHT

CONFIRMED BY CATL FACTORY CHECKLIST

J3.1 <-> J3.17	crimped jumper 'key'
J3.2 / J3.19	+24 V / 0 V from panel fuse BMSn
J3.6 / J3.23	NOT WIRED (no-connects) - fire release is Option B, via the FACP
J5.13 / J5.14	MCAN H/L to rack JX1_IN 9/10 (yel/grn twisted)
J5.29 / J5.30	pair A to ETH J902.6 / J902.5
J5.33 / J5.34	pair B to ETH J902.22/21, 60R crimped at MBMU
J902.1 / .13	ETH +24 V / 0 V, panel fuse 'ETH'
J902 9-12, 10-11	crimped jumpers (ETH-internal termination)
J902 RJ45	Ethernet to SW1 VLAN 30, Cat 6A shielded
ETH housing	bond to PE (CATL checklist)

THE 60 - OHM COMMISSIONING CHECK

Measure J902 pins 21/22: expect ~60 ohm. If ~30 ohm, the bus is over-

terminated: remove the crimped 60R (R1-R3). CATL ICD item - do not guess.

FIRE RELEASE = OPTION B (REV K DECISION) - NEVER WIRE BOTH

FACP TRIG > PSN-1000 > DS1 service disconnect > cabinet aerosol input (JX1 3/4).

B: Potter supervised circuits (the Carson way). Decide at IFC with CATL + AHJ.

CONNECTOR HARDWARE – EXACT PARTS ONLY

Plugs J5 / J3 / J902

360242-00175 / -00136 / -00078

CATL/Recodeal - ship with cabinets; verify at order



Rack JX1_In / JX2_Out

360242-00813 / -00814

REVL16P100A series: loop + aerosol + MCAN chain



JXH1 aux plug + leads

360242-00760 + AWG10 JONHON

A/B = chiller, C/D = rack BMS, ~230 VAC



Signal crimps

JAE M34S75C4F2 (0.5 mm²)

ALL CAN/COMM/24V conductors into MX34 plugs



Shield crimps

Molex M34S75C4F3 - DEDICATED

CAN shield only - different part than signal (CATL SOP)



Crimp quality here is the difference between a working CAN bus and a haunted one.

Follow the Molex MX34JAH SOP: twist 20-30 mm, single-point shield ground, runs under 30 m.

WIRE + CABLE SCHEDULE

C O L O R = C L A S S , O N E V E R Y D R A W I N G

4 8 0 V P O W E R	8 AWG THHN/MTW Cu; taps 14 AWG; SPD leads 10 AWG	R E D
A U X 3 8 0 Y / 2 2 0 V	AWG10 feeder + branches; CATL AWG10 at JXH1	O R A N G E
1 2 0 / 2 4 0 V C O N T R O L	14 AWG MTW	Y E L L O W
2 4 V D C + / C O M	16 AWG MTW	B L U E / C Y A N
R E L E A S I N G C K T S	FPL 14 AWG, one circuit per cabinet (TB14)	D K R E D
C A N B U S	AWG20 twisted shielded pair, <=30 m runs	T E A L
C R I M P J U M P E R S	AWG20 inside MX34 plugs	D K G R E Y
E T H E R N E T	Cat 6A shielded min (Cat 7 to ETH modules)	P U R P L E
F I R E C I R C U I T S	FPL 14-18 AWG per Potter / NFPA 72	D K R E D
C T L E A D S	18 AWG twisted	D K G R E E N
P E / G R O U N D	8 AWG Cu min; rack ground 50 mm2 (field)	G R E E N

All 174 nets belong to exactly one class - coverage is machine-checked at every build.

GROUNDING MAP + FIRST WEEK

ONE BOND PER SYSTEM - NO EXCEPTIONS

120/240 V system

N-G bond BD1 at the CPT secondary. The ONLY bond in this panel.

380Y/220 V aux

Bonded at the Kehua TX inside the PCS cabinet. Never add a second.

CAN shields

Single-point ground at SB1-3, panel end only. Floats at the cabinet.

ETH housings

Bond to PE (CATL checklist).

Rack grounds

50 mm² Cu, M8 at 10 N-m. Field scope, by others.

YOUR FIRST WEEK, IN ORDER

- 1 Trace one cabinet end-to-end on the one-page schematic: 24 V, CAN, pair A/B, aerosol, aux.
- 2 Run the Live Simulator - all six scenarios, slow motion.
- 3 Read the board register + Rev K disposition - especially R-01 and R-02.
- 4 Open the BOM beside the schematic: refs match 1:1 both ways.
- 5 Read the CATL external wiring diagram + checklist (page 15 derives from it).
- 6 Rebuild the drawings (buildG.py); watch 174/174 pass. Now you trust the pipeline.

WHAT GATES CONSTRUCTION

DO NOT LET ANYONE SKIP THESE



PE REVIEW + STAMP

nothing releases without it



SCCR CALCULATION

panel rating vs available fault current



RFI - 001 KEHUA

EPO contact sense, aux breaker ratings, N-G bond



RFI - 002 CATL / POTTER

JX1 aerosol pins, chiller nameplate, 24V loads



R1 - R3 60R

commissioning measurement decides (page 15)



KEHUA SUBMITTAL

aux breaker ratings + N-G bond confirmation



BATTALION VLAN

network plan sign-off



NFPA 72 BATTERY CALC

finalize with measured quiescent



WIRE NUMBERS

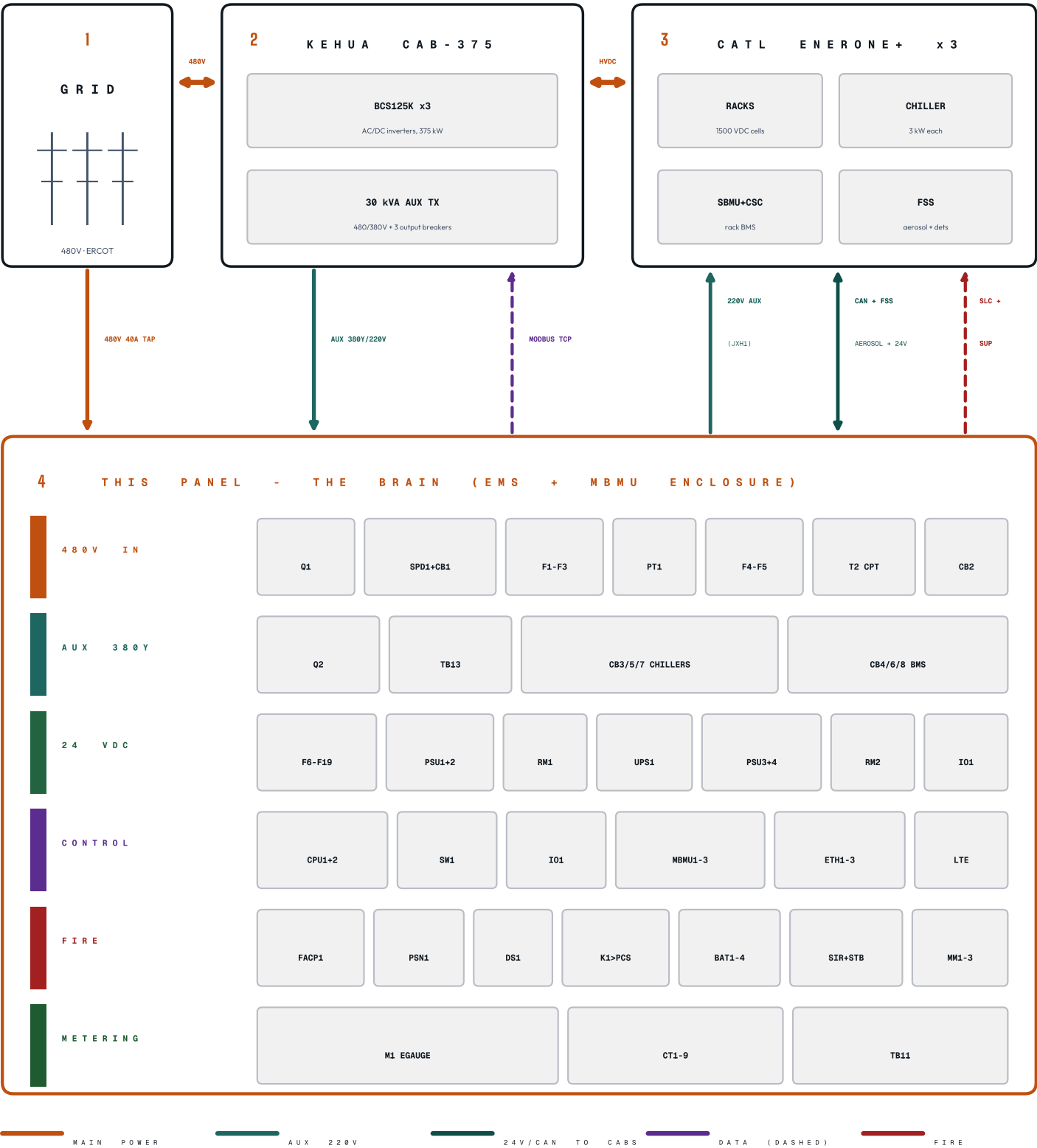
assigned at fab release

ABOUT THE QR CODES

Each code opens a search locked to that manufacturer's own site for the exact

model - stable even when vendors reshuffle URLs. Print the datasheets into the binder.

THE WHOLE SYSTEM – FOUR BOXES



Flow only - no wiring detail. For conductors, terminals and pins: the one-page ARCH E schematic.